

MATERIAL SAFETY DATA SHEET

(CLETHODIM 24% W/V EC)

SECTION 1. IDENTIFICATION OF PRODUCT

Common name:

Clethodim 24% W/V EC

Chemical name:

(5RS)-2-[(E)-1-[(2E)-3-chloroallyloxyimino] propyl]-5-[(2RS)-2-(ethylthio) propyl]-3-hydroxycyclohex-2-en-1-one

CAS No.: 99129-21-2

Manufacturer: SHANDONG CYNDA CHEMICAL CO., LTD.

Address: ECONOMIC DEVELOPMENT AREA, BOXING COUNTY, BINGZHOU CITY,
SHANDONG PROVINCE, P. R. CHINA.

Emergency phone number: +86-543-2320028

Recommended use: Herbicide

SECTION 2. COMPOSITION INFORMATION ON INGREDIENTS

COMPOSITION:

<u>Clethodim 24% W/V EC</u>	<u>W/V</u>
Clethodim	24%
<u>Inert ingredients</u>	<u>76%</u>
Total	100%

SECTION 3. IDENTITY

Avoid direct physical contact!

Mild irritant to skin and mild transient irritant to eye.

SECTION 4. FIRST AID

If poisoning occurs, immediately contact a doctor or Poisons Information Centre, and follow the advice given. Show this Material Safety Data Sheet to the doctor. Have the product container or label with you when calling a poison control center or doctor, or going for treatment,

EYES:

Hold eye open and rinse slowly and gently with water for 15-20 minutes.

Remove contact lenses, if present, after the first 5 minutes, then continue rinsing eye.

Call a poison control center or doctor for treatment advice.

SKIN:

Take off contaminated clothing.

Rinse skin immediately with plenty of water for 15-20 minutes.

Call a poison control center or doctor for treatment advice.

INGESTION:

Call a poison control center or doctor immediately for treatment advice.

Have a person sip a glass of water if able to swallow.

Do not induce vomiting unless told to do so by the poison control center or doctor.

Do not give anything by mouth to an unconscious person.

SECTION 5. FIRE AND EXPLOSION HAZARD DATA

Flash point : 144°- 150°F (Technical)

Extinguishing media: Carbon dioxide, dry chemical, foam, water fog.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Use the personal protective equipment listed in Chapter 8. Prevent entry into drains, waters or soil. Take up spilled product with absorbent material (e.g. sawdust, peat, chemical binder). Fill materials taken up into closable container. To clean the floor and all objects contaminated by this material, use damp cloth. Also place used cleaning materials

into closable receptacles.

SECTION 7. HANDLING AND STORAGE

INFORMATION ON SAFE HANDLING:

Suitable container materials: HDPE (high density polyethylene).

When using open containers, use local exhaust ventilation to prevent vapours from spreading. Make provision for product and fire-fighting water to be retained.

INFORMATION ON STORAGE:

See Chapter 15 for storage regulations. Observe the rules contained in the VCI concept for separate/common storage. To maintain quality, store in a dry place and protect from temperatures below -10 and above 40 C°. Store so that unauthorized persons do not have access. Keep away from food, drink and animal feeding stuffs.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Personal Protective Equipment:

Eye Protection:

This product does not cause significant eye irritation or eye toxicity requiring special protection. Use good handling practices to avoid eye contact.

Skin Protection:

Wear appropriate protective clothing and waterproof gloves to prevent skin contact. Applicators and other handlers must wear: long-sleeved shirt, long pants and shoes with socks. Wash hands and contaminated skin thoroughly after handling.

Follow manufacturer's instructions for cleaning/maintaining PPE. If no such instructions for washables, use detergent and hot water. Keep and wash PPE separately from other laundry.

Attention! Repeated or prolonged contact may cause allergic skin reaction in some people.

Respiratory Protection: Avoid mist. This product is not likely to pose an airborne exposure concern when handled and used in accordance with label instructions.

Ventilation:

No special precautions recommended when handled in accordance with label instructions.

Exposure Guidelines:

<u>Exposure Limits</u>	<u>OSHA PEL</u>	<u>ACGIH TLV</u>
MICRO-TECH(R)	None established	None established
Clethodim	None established	None established

SECTION 9. PHYSICAL AND CHEMICAL CHARACTERISTICS

Appearance : yellowish to orange homogenous liquid.

Odor : Solvent odor

Relative density : 0.96 ± 0.02

pH value : 3.0-7.0

SECTION 10. REACTIVITY DATA

Material is stable under recommended storage conditions. Material is unstable at extreme pH's, temperature and upon exposure to UV light.

Avoid contact with oxidizing materials such as chlorates, nitrates, peroxides, etc. In a fire, carbon dioxide, water vapor and oxides of nitrogen, sulfur may be formed. Incomplete combustion can produce carbon monoxide.

SECTION 11. TOXICOLOGICAL INFORMATION

Acute oral LD₅₀ in rats: >5000 mg/kg body weight

Acute percutaneous LD₅₀: >2150mg/kg

Do not cause skin irritation in rabbits, slight-irritation to eye.

SECTION 12. ECOLOGICAL INFORMATION

Birds: Oral LD₅₀ for bobwhite quail >2000 mg/kg. Dietary LC₅₀ for mallard ducks >6000 mg/kg.

Fish: LC₅₀ (96 h) for rainbow trout 67, bluegill sunfish >120 mg/l.

Daphnia: LC₅₀ (48 h) >120 mg/l; NOEC 60 mg/l.

Algae: EC₅₀ (5 d) for fresh-water algae 57.8 mg/l.

Bees: LD₅₀ (contact) >100 mg/bee.

Worms: LC₅₀ for worms 454 mg/kg soil; NOEL 316 mg/kg soil.

SECTION 13. DISPOSAL EMPTY CONTAINERS

All drums may be returned to the manufacturer provided they are clean, serviceable, have bungs and plugs in place and are free of surface chemical residues. Unserviceable drums

should be triple rinsed with water, have the liner removed (where applicable) and be crushed to prevent re-use. Disposal of drums and liners may be permitted into a tip after approval from the local authority.

SECTION 14. TRANSPORT INFORMATION

D.O.T. SHIPPING: Compounds, weed killing, liquid, non-regulated.

TECHNICAL SHIPPING NAME: Clethodim 24% W/V EC

D.O.T. HAZARD CLASS: NA

SECTION 15. REGULATORY INFORMATION

All pesticides are governed federal act. Therefore, the regulations are pertinent only when handled outside of the normal use and applications of pesticide. This includes waste streams resulting from manufacturing/formulation facilities, spills or misuse of products, and storage of large quantities of products containing hazardous or extremely hazardous substances.

SECTION 16. OTHER INFORMATION

The above information is believed to be correct on the date it is published and must not be considered all inclusive. The information has been obtained only by a search of available literature and is only a guide for handling the chemicals.

OSHA regulations require that if other hazards become evident, an upgraded MSDS must be made available to the employee within three months.

Responsibility for updates lies with the employer. Persons not specifically and properly trained should not handle this chemical or its container. This MSDS is provided without any warranty expressed or implied, including merchantability or fitness for any particular purpose.